

PrivSTD: Differentially Private Spatio-temporal Trajectory Density Data Publication

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ABSTRACT

Spatio-temporal trajectory density data is used widely in, e.g., urban analytics, mobility, and epidemiology. While differential privacy guarantees are important enablers of the release and use of such data, it is challenging to offer privacy guarantees for high-resolution density data. Specifically, existing approaches inject noise into the spatial domain, failing to preserve inherent spatio-temporal correlations and suffering from severely reduced utility at fine granularities. We propose PrivSTD, a novel framework for differentially private release of spatio-temporal trajectory density data. PrivSTD leverages the Discrete Cosine Transform to project density data into the frequency domain, where spatial correlations and temporal smoothness are naturally captured by low-frequency components. To suppress noise-dominated frequencies, a Benjamini–Hochberg FDR-based adaptive truncation mechanism is introduced that preserves statistically significant structures without additional privacy cost. Furthermore, PrivSTD employs a control variate-enhanced Recorruped-to-Recorruped denoising model to reconstruct high-quality density data without access to clean ground truth data. An experimental study shows that PrivSTD is capable outperforming existing methods, achieving 1.12X–53.89X (6.12X on average) lower error across all datasets considered.

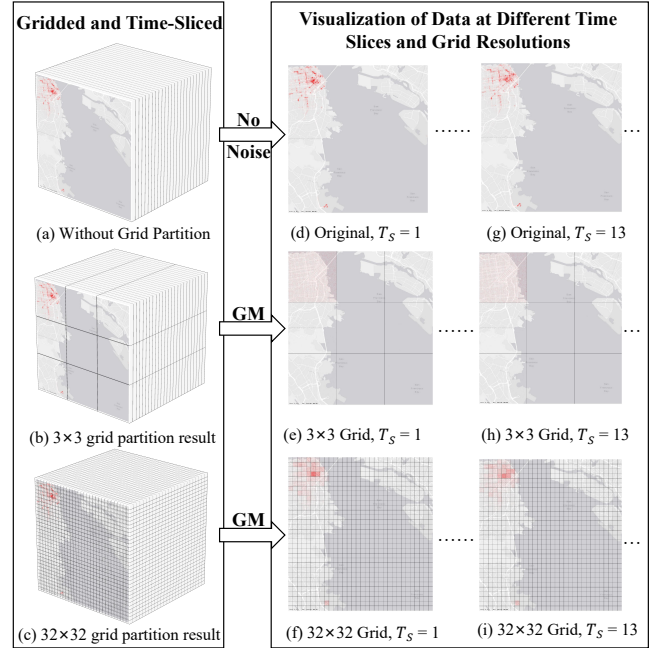


Figure 1: Motivating Example.

PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/YeShuzhan/PrivSTD>.

1 INTRODUCTION

Large-scale spatiotemporal trajectory data, generated by location-sensing devices such as smartphones and GPS sensors, have emerged as indispensable resources across diverse domains, including urban planning [24, 52], epidemiology [11, 34, 35], and human mobility studies [16, 57].

For instance, initiatives such as “Data for Good” [21] have released public spatiotemporal density datasets to support epidemic tracking and response [48], typically in the form of counts assigned to the cells in fine-grained spatio-temporal grids [15, 26]. To offer privacy protection while preserving analytical utility, *differential privacy* (DP) has been proposed, furnishing stringent mathematical assurances that curtail an adversary’s ability to infer the inclusion of any single data element [18]. DP enjoys widespread adoption by both industrial and governmental entities, including by U.S. Census Bureau [28], Uber [27], and LinkedIn [43].

Methods for privacy-preserving release of spatiotemporal trajectory density data employ histograms [22, 40, 41, 54] or attribute distribution modeling paradigms [10, 33, 53], complemented by post-hoc denoising strategies to refine disseminated outputs. The state-of-the-art methods VDR, predicated on the analogous distributional modalities between spatiotemporal trajectory density data and image data, deploy the VQ-VAE model as a post-processing mechanism and employ multi-resolution learning for the release of spatiotemporal trajectory density data [4]. However, existing methods do not address three fundamental challenges, which we illustrate using the real-world CABS_SF dataset in Figure 1. To address these challenges, we propose PrivSTD, an event-level framework for high-precision spatiotemporal trajectory density release.

Challenge I: How to reconcile grid resolution and data utility under differential privacy? Density analyses of spatiotemporal data, time slicing and grid partitioning. The grid resolution plays a pivotal role in capturing both local patterns and global mobility trends. As illustrated in Figure 1, a low resolution (3×3) yields coarse distributions, whereas increasing to a higher resolution (32×32) enables reconstructions that approximate the original data. However, this

increased resolution drastically dilutes trajectory counts per cell, creating a sparsity issue where DP noise overwhelms the weak signal. Consequently, direct grid-based perturbations suffer severe utility erosion as resolution increases. To address this resolution–noise trade-off, PrivSTD utilizes the Discrete Cosine Transform (DCT) to map density data into the frequency domain. Unlike the spatial domain, where the information is dispersed, DCT maps the data features into a compact set of low-frequency coefficients. This transformation decouples data utility from grid granularity, allowing the preservation of core features even under high-resolution settings.

Challenge II: How to capture spatial and temporal correlations in trajectory density data? As shown in Figure 1(a) and 1(b), hotspots in both the first time slice ($T_S = 1$) and the 13th time slice ($T_S = 13$) concentrate around similar regions, despite a nearly one-year interval, underscoring the strong spatial autocorrelation (adjacent grid cells exhibit similar densities due to road networks and continuous movement) and temporal smoothness (human mobility follows predictable daily/weekly rhythms) inherent in trajectory density data. However, existing methods often overlook these intrinsic characteristics. Specifically, nearly all current differential privacy mechanisms operate by independently perturbing the trajectory point count within each individual grid cell, resulting in fragmented spatiotemporal patterns and diminished data utility. To overcome this, PrivSTD capitalizes on the persistence of patterns across time. Specifically, it aggregates the spectral energy of corresponding frequencies across all time slices to derive a cumulative statistic, denoted as $S(u, v)$. Since persistent spatiotemporal patterns (e.g., recurring hotspots and smooth spatial/temporal variations) appear as stable high-energy components in $S(u, v)$ distinct from random noise, we employ the Benjamini-Hochberg False Discovery Rate (BH-FDR) on this statistic. This process aims to accurately identify and retain significant coefficients.

Challenge III: How to achieve effective denoising without clean ground truth data under DP constraints? Under differential privacy constraints, supervised denoising is infeasible due to the lack of clean ground truth data, forcing methods to rely on unsupervised objectives. However, existing unsupervised approaches (e.g., VQ-VAE in VDR, plain R2R, or score matching) often collapse into near-identity mappings that overfit to structured and heteroscedastic noise, especially in high-resolution spatiotemporal settings. High dimensionality and varying noise scales across frequency bands further exacerbate training instability. To address this, we introduce the Control Variate-Enhanced Recorrupted-to-Recorrupted (CV-R2R) denoising model, which transforms unsupervised denoising into a variance-reduced equivalent of supervised training without observing any clean data. By incorporating optimally weighted control variates, combined with weighted Generalized Least Squares (GLS) and control variate-aware Stochastic Polyak Step-size (CV-SPS), CV-R2R ensures stable convergence and near-supervised performance even amid severe noise heteroscedasticity.

Our contributions are summarized as follows.

- We propose PrivSTD, an efficient framework for releasing spatiotemporal trajectory density data that generates synthetic densities satisfying event-level differential privacy guarantees while preserving essential spatiotemporal features (Section 4).
- We combine a DCT-based mechanism with BH-FDR-controlled adaptive frequency-domain truncation, alleviates noise explosion in high-resolution scenarios and captures spatiotemporal characteristics across time slices (Section 5).
- We present CV-R2R, an unsupervised denoising model designed for heteroscedastic noise environments, which is able to achieve statistical equivalence in expectation to supervised denoising, enabling stable and high-utility reconstruction without access to clean data (Section 6).
- We report on extensive experiments involving three real-world spatiotemporal trajectory density datasets, offering insight into PrivSTD’s effectiveness and efficiency through a variety of evaluation metrics (Section 7).

2 RELATED WORK

Private Data Release. Since a spatiotemporal density data grid is a special case of multidimensional histogram, our work relates closely to differentially private histogram publishing. Existing solutions can be categorized into optimization-based, synthesis-based, and learning-based methods. Optimization-based methods, e.g., AHP [54] and MWEM [22], reduce noise through adaptive grouping or iterative refinement. Synthesis-based methods, e.g., PrivBayes [53], construct a generative model to approximate the data distribution, such as PrivBayes [53]. Learning-based methods model complex spatiotemporal dependencies, including methods that decompose data into independent temporal series [2], apply deep generative denoising [4], or approximate query answers via neural databases [51]. However, these methods either restrict query expressiveness, require predefined workloads, or remain sensitive to high-dimensional noise. In particular, state-of-the-art methods like VDR [4] still suffer from considerable utility loss in high-resolution spatiotemporal density data release due to amplified noise and limited exploitation of cross-temporal correlations.

Denoising Techniques. Most deep learning-based denoising methods rely on clean–noisy or multiple noisy observations [25, 31, 32, 37], assumptions that are incompatible with differential privacy due to repeated accesses to the same data. While some unsupervised techniques can mitigate low-level noise [37, 56], denoising under DP constraints remains under-explored. Notably, recent advances [4] have begun to address denoising under DP constraints by employing regularized representation learning using variational autoencoders, specifically designed to handle spatiotemporal data patterns while maintaining formal privacy guarantees. Nevertheless, existing unsupervised approaches often degenerate into near-identity mappings that overfit to structured and heteroscedastic noise, particularly in high-resolution settings [12, 30].

Differentially Private Machine Learning. Machine learning models are vulnerable to leaking training data information [23, 44]. DP has been applied to empirical risk minimization [9, 29] and deep neural networks [1, 47], typically by adding noise to model outputs [49], introducing regularization terms [9, 29], or perturbing gradients [1]. In contrast, VDR [4] sanitizes training data, decoupling privacy protection from learning algorithms. These methods generally achieve higher utility on high-dimensional data and avoid overfitting issues commonly seen in gradient-based mechanisms.

Table 1: Summary of Notation

ϵ	Privacy budget
δ	Probability of failure
Z	Gaussian noise
H	Grid size (column dimension)
W	Grid size (row dimension)
U	Frequency bandwidth (row-wise)
V	Frequency bandwidth (column-wise)
Θ	2D DCT coefficient matrix
X	Grid-based count/density matrix
C_x	Orthonormal DCT basis matrix (row dimension)
C_y	Orthonormal DCT basis matrix (column dimension)
$s_x(i)$	Prefix energy along DCT row basis
$s_y(j)$	Prefix energy along DCT column basis
$S(u, v)$	Chi-square test statistic
L	Covariance factor
C	Control variate
$L(\theta)$	Loss function of R2R
$\mathcal{L}(\theta)$	Loss function of CV-R2R
W	Precision weight matrix

3 PRELIMINARIES

3.1 Differential Privacy

Differential Privacy (DP) [18] is a framework designed for scenarios involving a trusted data curator. Intuitively, DP ensures that the contribution of any single data element to a final output is limited. In its original formulation, DP is characterized by a single privacy parameter ϵ . Over the past two decades scholars have argued that this definition is often too stringent for practical applications and have proposed various relaxations. Among these, the most widely adopted is *approximate DP* [18], which introduces an additional parameter δ .

DEFINITION 1. (ϵ, δ) -Differential Privacy $((\epsilon, \delta)$ -DP). An algorithm $\mathcal{A}$ satisfies ϵ -differential privacy, where $\epsilon > 0$ and $\delta \geq 0$, if and only if for any two neighboring datasets D and D' , we have $\forall T \subseteq \text{Range}(\mathcal{A}) : \Pr[\mathcal{A}(D) \in T] \leq e^\epsilon \cdot \Pr[\mathcal{A}(D') \in T] + \delta$.

We consider two datasets D and D' to be **event-level** neighbors, denoted as $D \simeq D'$, when D' is derived from D by adding or removing a single data record.

Gaussian Mechanism (GM) [55]. Given a query/aggregation function f , the GM defines a randomized algorithm $\mathcal{A}_f$ (below) that satisfies (ϵ, δ) -DP by adding Gaussian noise to $f(D)$. The magnitude of the noise depends on GS_f , i.e., *global sensitivity*,

$$GS_f = \max_{D \simeq D'} \|f(D) - f(D')\|_2,$$

where f denotes the aggregation function, and D or D' are neighboring dataset. When f outputs a scalar, the Gaussian mechanism $\mathcal{A}$ is as:

$$\mathcal{A}_f(D) = f(D) + \mathcal{N}(0, \sigma^2),$$

where $\mathcal{N}(0, \sigma^2)$ denotes a random variable sampled from the Gaussian distribution with scale parameter $\sigma^2 \geq \frac{\sqrt{2 \ln(1.25/\delta)} \Delta f}{\epsilon}$. When f outputs a vector, $\mathcal{A}$ adds independent samples of $\mathcal{N}(0, \sigma^2)$ to each element of the vector.

3.2 Problem Formulation

We consider a dataset D of user location reports (lon, lat, time, id) and aim to release a corresponding high-resolution spatiotemporal distribution while offering differential privacy for use in analyses. We discretize space and time into a three-dimensional histogram $H \in \mathbb{R}^{M \times M \times T}$, and report a user location count for each cell.

We design a mechanism $\mathcal{M}$ that takes H as input and outputs $\hat{H} = \mathcal{M}(H)$ satisfying (ϵ, δ) -DP, enabling task-agnostic analyses without any predefined assumptions. Our objective is to maximize the utility of $\hat{H}$ as an approximation of H by preserving spatial correlations and temporal continuity, which are important for common analyses (e.g., aggregation and pattern discovery). To evaluate the effectiveness of the proposed mechanism, we consider several representative tasks and corresponding utility metrics in Section 7. These tasks are chosen for evaluation purposes and do not constrain or specialize the design of our mechanism, which remains task-agnostic.

4 OVERVIEW OF PRIVSTD

As illustrated in Figure 2, the PrivSTD method is designed for differentially private release of high-resolution spatiotemporal trajectory density data. It consists of two components, enabling *Private Frequency Estimation* and *Self-Supervised Denoising*.

Private Frequency Estimation. PrivSTD first discretizes trajectory data to a spatio-temporal grid. For each time slice, a 2D Discrete Cosine Transform (DCT) is applied to project spatial densities into the frequency domain, where spatial correlations are captured by low-frequency coefficients. Using Gaussian mechanism (GM) noise is injected into the frequency coefficients to guarantee event-level differential privacy. To suppress noise-dominated components, PrivSTD adopts a Benjamini-Hochberg False Discovery Rate (BH-FDR) based truncation strategy and applies an inverse DCT (IDCT) to reconstruct noisy preliminary density estimates. The retained DCT bases further induce a covariance matrix, which is used in the subsequent denoising stage.

Unsupervised Denoising. Although the frequency-domain mechanism preserves the dominant spatiotemporal structures, the inverse-DCT reconstructed private density estimate still contains structured and heteroscedastic differential privacy noise. To recover high-quality results without access to the original data, PrivSTD performs a low-rank factorization of the previously constructed covariance matrix to derive a *covariance factor* L , and it then samples two independent noise vectors z^a and z^b from a shared Gaussian noise distribution. Based on z^a , z^b , and a scaling parameter α , PrivSTD applies linear transformations to construct a pair of statistically consistent yet independent inputs y^+ and y^- .

Next, PrivSTD feeds y^+ into a denoising network to obtain the prediction $\mathcal{F}_\theta(y^+)$. Using y^- as an unbiased supervision signal, the method employs a recorrputed-to-recorrputed learning, where the discrepancy between $\mathcal{F}_\theta(y^+)$ and y^- serves as the basic training objective. Building upon this objective, PrivSTD further incorporates a *precision weight matrix* W induced by the covariance structure, together with a control variate C_W derived from W , to refine the training objective and construct a control-variate augmented loss function $\hat{\mathcal{L}}_{cv, W}(\theta)$ (denoted as $\hat{\mathcal{L}}(\theta)$ for brevity), which effectively reduces the variance of stochastic gradients while preserving unbiased gradient estimation.

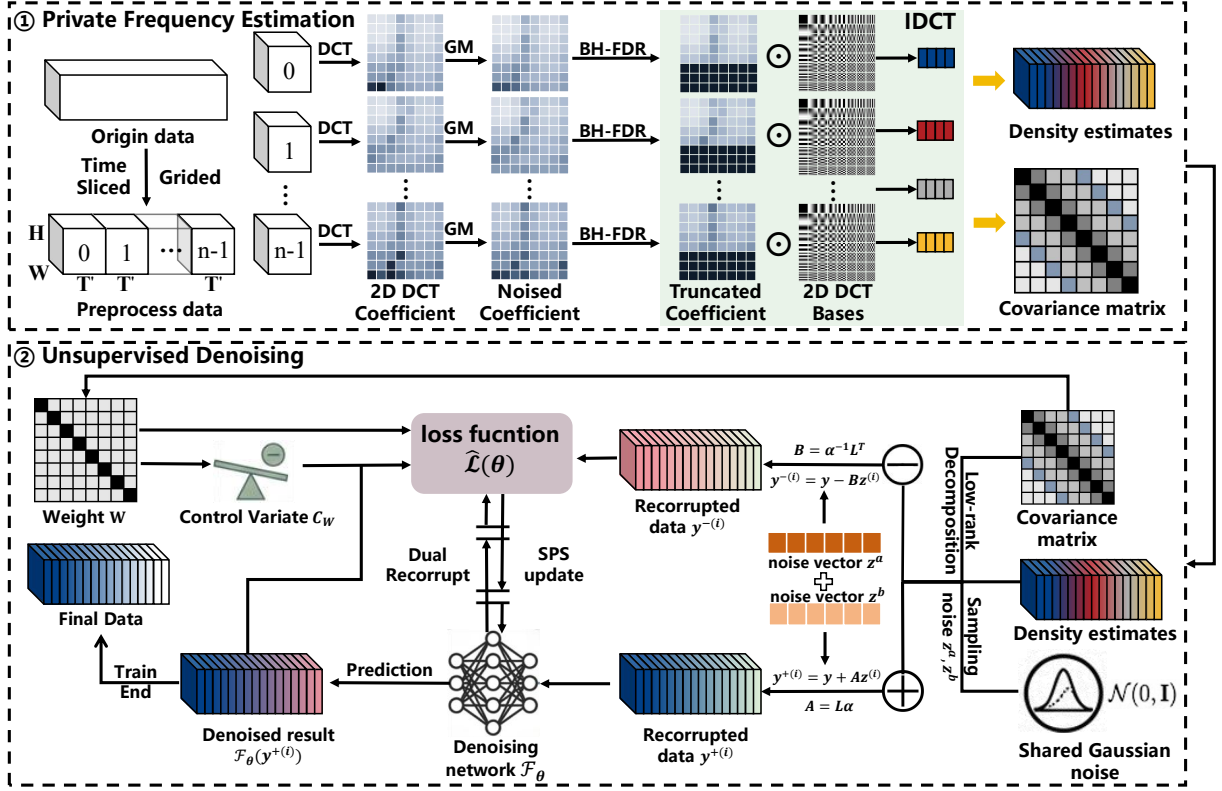


Figure 2: Framework of PrivSTD

During training, the dual-sample structure enables online estimation of stochastic gradient variability using two independent noise realizations each iteration. This allows PrivSTD to perform a *Stochastic Polyak Step-size* (SPS) update at every parameter update, adaptively adjusting the learning rate according to both loss suboptimality and noise-induced uncertainty.

5 PRIVATE FREQUENCY ESTIMATION

Private frequency estimation releases dominant spectral components of spatio-temporal density grid cells while satisfying differential privacy. Naïve cell-wise perturbation ignores spatial smoothness, dispersing variance across frequencies and degrading utility [19, 46]. We therefore estimate spectra directly by using with the *Discrete Cosine Transform Mechanism* (DCTM), which perturbs decorrelated coefficients with bounded ℓ_2 sensitivity and reconstructs sanitized densities via inverse transformation.

5.1 The Discrete Cosine Transform

The Discrete Cosine Transform (DCT) is a linear, invertible orthogonal transform with strong energy compaction [3, 42]. We use the DCT-II basis to represent grid cells as sums of cosine modes at increasing spatial frequencies. For a one-dimensional signal of length N , the DCT-II is defined as follows:

$$Y_k = \sum_{n=0}^{N-1} x_n \cos \left[\frac{\pi}{N} \left(n + \frac{1}{2} \right) k \right], \quad k = 0, \dots, N-1,$$

with standard orthonormal scaling.

For a grid count matrix $X \in \mathbb{R}^{H \times W}$, the 2D DCT-II is computed separately:

$$\Theta_{u,v} = \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} X_{i,j} C_x[i, u] C_y[j, v], \quad u = 0, \dots, H-1; v = 0, \dots, W-1, \quad (1)$$

or in matrix form:

$$\Theta = C_x^T X C_y,$$

where $C_x \in \mathbb{R}^{H \times H}$ and $C_y \in \mathbb{R}^{W \times W}$ are orthogonal cosine bases:

$$C_x[i, u] = \begin{cases} \frac{1}{\sqrt{H}}, & u = 0 \\ \sqrt{\frac{2}{H}} \cos\left(\frac{\pi(2i+1)u}{2H}\right), & u \geq 1 \end{cases},$$

$$C_y[j, v] = \begin{cases} \frac{1}{\sqrt{W}}, & v = 0 \\ \sqrt{\frac{2}{W}} \cos\left(\frac{\pi(2j+1)v}{2W}\right), & v \geq 1 \end{cases}.$$

Orthogonality ($C_x^T C_x = I_H$, $C_y^T C_y = I_W$) implies Parseval preservation $\|X\|_F^2 = \|\Theta\|_F^2$ and decorrelates grid cells, concentrating energy in low-frequency coefficients near $\Theta_{0,0}$ [3]. This is well matched to trajectory densities with smooth spatial variation, and reconstruction uses the inverse DCT:

$$X = C_x \Theta C_y^T.$$

5.2 DCT under Differential Privacy

As a concrete instantiation of private frequency estimation on grid-structured data, DCTM perturbs DCT coefficients while bounding individual contributions. The DCTM treats the flattened coefficient

vector as a query function $f(D) = \text{vec}(\Theta)$, where D is the dataset of location reports aggregated into X . For event-level adjacency (neighboring datasets $D \sim D'$ differ by adding/removing one report, altering one grid cell by ± 1), the global ℓ_2 sensitivity Δ_2 quantifies the maximum change in f .

Next, let $C_x \in \mathbb{R}^{m \times m}$ and $C_y \in \mathbb{R}^{n \times n}$ be orthonormal DCT matrices and define partial energies $s_x(i; H)$ and $s_y(j; W)$ for bands $H \leq m$ and $W \leq n$ by

$$\begin{aligned} s_x(i; H) &= \sum_{u < H} C_x[i, u]^2, \\ s_y(j; W) &= \sum_{v < W} C_y[j, v]^2, \\ w_{ij}(H, W) &= s_x(i; H) s_y(j; W). \end{aligned}$$

We use zero-based frequency indices here; adopting a different convention only reindexes the same sums.

LEMMA 1. *For all i, j, U, V , we have $0 \leq s_x(i; U), s_y(j; V) \leq 1$ with these quantities nondecreasing in U and V and attaining 1 at $U = H$ and $V = W$, further $\sum_i s_x(i; U) = U$ and $\sum_j s_y(j; V) = V$.*

PROOF. From $C_x^\top C_x = I_H$ we get $C_x C_x^\top = I_H$, so the i th diagonal entry yields $\sum_u C_x[i, u]^2 = 1$. Since $s_x(i; U)$ is a partial sum of nonnegative terms, $s_x(i; U) \leq 1$ and is nondecreasing in U , with equality at $U = H$. Repeat verbatim for C_y to obtain the claims for $s_y(j; V)$. $\square$

THEOREM 1. *The global DCT ℓ_2 sensitivity $\Delta_2(H, W) = \sqrt{\max_i s_x(i; H) \cdot \max_j s_y(j; W)} \leq 1$.*

PROOF. Consider the event-level adjacency notion, where adjacent datasets $D \sim D'$ differ by the addition/removal of a single count in one cell location. Let the query keep only the $H \times W$ low-frequency 2D-DCT coefficients for each time frame. Let $X^{(t)} \in \mathbb{R}^{m \times n}$ be the count matrix at time t and $x^{(t)} = \text{vec}(X^{(t)}) \in \mathbb{R}^{mn}$. The orthonormal 2D DCT of a frame is

$$\Theta^{(t)} = C_x^\top X^{(t)} C_y, \quad \Theta_{u,v}^{(t)} = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} C_x[i, u] X_{i,j}^{(t)} C_y[j, v].$$

We apply stacking (e.g., column-major) to obtain a vector:

$$g^{(t)}(D) = \text{vec}(\Theta_{0:H, 0:W}^{(t)}) = (C_y^{(:, 0:W)} \otimes C_x^{(:, 0:H)})^\top x^{(t)} \in \mathbb{R}^K, K = HW.$$

For adjacent datasets D and D' , a unique (i_0, j_0, t_0) exists such that $x^{(t_0)} - x'^{(t_0)} = \pm e_{i_0 j_0}$ (the canonical basis vector of $\mathbb{R}^{mn}$ at grid cell (i_0, j_0)), while $x^{(t)} = x'^{(t)}$ for $t \neq t_0$. Hence $g^{(t)}(D) - g^{(t)}(D') = 0$ for $t \neq t_0$, further at $t = t_0$, we have:

$$g^{(t_0)}(D) - g^{(t_0)}(D') = M^\top (x^{(t_0)} - x'^{(t_0)}) = \pm M^\top e_{i_0 j_0}.$$

By the entrywise expression for $\Theta^{(t)}$, each kept coefficient changes as follows.

$$\Theta_{u,v}^{(t_0)}(D) - \Theta_{u,v}^{(t_0)}(D') = \pm C_x[i_0, u] C_y[j_0, v], 0 \leq u < H, 0 \leq v < W.$$

Therefore:

$$\begin{aligned} \|f(D) - f(D')\|_2^2 &= \|g^{(t_0)}(D) - g^{(t_0)}(D')\|_2^2 \\ &= \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} (C_x[i_0, u] C_y[j_0, v])^2 \\ &= s_x(i_0; H) s_y(j_0; W). \end{aligned}$$

Taking the worst-case (i_0, j_0) over all grid cells yields the maximum change in ℓ_2 norm between adjacent datasets. Therefore, we have:

$$\Delta_2(H, W) = \max_{D \sim D'} \|f(D) - f(D')\|_2 = \sqrt{\max_i s_x(i; H) \cdot \max_j s_y(j; W)}. \quad (2)$$

Based on Lemma 1, the upper bound of $\Delta_2(H, W)$ is 1. $\square$

DCTM perturbs coefficients by $\Theta_{\text{noisy}} = \Theta + Z$ with i.i.d. $Z_{uv} \sim \mathcal{N}(0, \sigma^2)$, and reconstructs $\tilde{X} = C_x \Theta_{\text{noisy}} C_y^\top$. The Gaussian scale for (ϵ, δ) -DP is

$$\sigma \geq \frac{\Delta_2 \sqrt{2 \ln(1.25/\delta)}}{\epsilon} = \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}. \quad (3)$$

THEOREM 2. *The Discrete Cosine Transform Mechanism satisfies (ϵ, δ) -DP.*

PROOF. Let $\mathcal{M}$ denote the DCTM, which computes $f(D)$ and injects Gaussian noise with variance σ^2 calibrated as in Equation (3). For neighboring datasets D and D' , the outputs $\mathcal{M}(D)$ and $\mathcal{M}(D')$ are distributed as multivariate Gaussians with means $f(D)$ and $f(D')$, respectively, and they have identical covariance $\sigma^2 I$.

The privacy guarantee follows from the standard analysis of the Gaussian mechanism for ℓ_2 -sensitivity [20]. Specifically, the probability density ratio is bounded by considering the worst-case shift in means, which is at most $\Delta_2 = 1$. The tail bound for Gaussians ensures that for any subset S of outputs and any $\delta > 0$,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta,$$

when $\sigma \geq \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}$. This holds because the divergence between two Gaussians separated by distance Δ_2 is controlled by the calibration, with the factor 1.25 arising from a tight union bound over possible failure events [19]. $\square$

The inverse DCT is employed for post-processing and incurs no additional privacy loss [20]. By orthogonality, noise energy is preserved under the transform; equivalently, reconstruction noise remains Gaussian with variance σ^2 per cell up to an orthogonal mixing.

5.3 BH-FDR Truncation

DP injects Gaussian noise across the spectrum, making high-frequency coefficients largely noise-dominated; due to the global support of DCT bases, such noise propagates throughout the spatial domain after inversion [36]. Frequency-domain truncation mitigates this effect, but fixed bandwidths are not adaptive to dataset spectra or privacy noise levels [17].

We therefore apply BH-FDR as a post-processing step on DP spectra to adaptively select an effective bandwidth without additional privacy cost.

FDR is defined as follows

$$\text{FDR} = \mathbb{E} \left[\frac{V}{\max(R, 1)} \right],$$

where V is the number of false rejections and R is the total rejections [6]. We use BH-FDR because it is less conservative than FWER while controlling the expected false-discovery proportion at high dimensions [45]. The BH procedure sorts p -values $p_{(1)} \leq \dots \leq p_{(m)}$ and selects the largest k such that $p_{(k)} \leq (k/m)q$, rejecting all

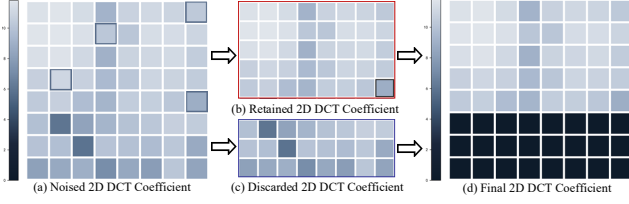


Figure 3: BH-FDR Example

$p \leq p_{(k)}$; it controls FDR under independence and admits correlated-test extensions [7].

In our DP-DCT framework:

$$\Theta_{\text{noisy}}^{\text{full}} = \Theta_{\text{true}} + Z, \quad Z_{uv} \sim \mathcal{N}(0, \sigma^2), \quad \sigma = \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}.$$

To leverage temporal redundancy across T frames, define

$$S(u, v) = \sum_{t=1}^T \frac{(\Theta_{\text{noisy}}^{(t)}[u, v])^2}{\sigma^2}.$$

Under the null hypothesis, $S(u, v) \sim \chi_T^2$, yielding

$$p(u, v) = 1 - F_{\chi_T^2}(S(u, v)),$$

with $F_{\chi_T^2}$ the chi-squared CDF [8]. Applying BH to the ($m = HW$) p-values yields a set $\mathcal{S}$ of frequency coordinates that reject the noise-only null hypothesis (at FDR level q), i.e., coefficients whose energy is unlikely to be explained by DP noise alone.

Figure 3 illustrates this process. In panel a, all two-dimensional DCT basis functions are displayed with noise added. BH-FDR identifies those coordinates exhibiting statistically significant evidence of nonzero signal; in the example, four such basis elements are detected (the bases surrounded by black box). The bottom right corner box represents the highest-frequency significant coordinate and is treated as the outer boundary of the effective frequency band. Comparing panels (a)–(c), we observe a clear reduction in noise energy: the full noisy spectrum contains approximately $64\sigma^2$ of noise energy, whereas the BH-truncated region retains only about $40\sigma^2$. This reduction shows how removing noise-dominated high-frequency components effectively limits the propagation of noise in the spatial domain.

Rather than retaining only the pointwise significant set $\mathcal{S}$, we adopt a rectangularized low-pass region to avoid fragmented masks that create spatial artifacts. Under low-frequency dominance and spectral connectivity [39], if $(u, v) \in \mathcal{S}$ with large u or v , the corresponding coefficients of lower frequencies typically carry an informative structure. However, these may fail BH significance due to DP noise. We therefore define $U^* = \max_{(u,v) \in \mathcal{S}} u$ and $V^* = \max_{(u,v) \in \mathcal{S}} v$, and then retain $[0 : U^*] \times [0 : V^*]$ as the final bandwidth. Formally, we define:

$$U^* = \max_{(u,v) \in \mathcal{S}} u, \quad V^* = \max_{(u,v) \in \mathcal{S}} v,$$

and retain the region $[0 : U^*] \times [0 : V^*]$ as the final frequency bandwidth.

The truncated coefficients are then given as follows.

$$\Theta_{\text{noisy}}^{\text{trunc}}[u, v] = \begin{cases} \Theta_{\text{noisy}}^{\text{full}}[u, v], & (u, v) \in [0 : U^*] \times [0 : V^*], \\ 0, & \text{otherwise,} \end{cases}$$

Reconstruction then proceeds via as follows.

$$\tilde{X} = C_x \Theta_{\text{noisy}}^{\text{trunc}} C_y^T.$$

BH-FDR truncation reduces high-frequency noise propagation; by Parseval lemma [42], the expected post-truncation noise energy is $\mathbb{E}[\|\tilde{E}\|_F^2] = U^* V^* \sigma^2$, versus $HW \sigma^2$ without truncation. Meanwhile, FDR control helps preserve coefficients carrying genuine signal energy and mitigates over-aggressive truncation bias.

For dependence tests, we may use the Benjamini–Yekutieli adjustment $q \leftarrow q / \sum_{i=1}^m 1/i$ to control FDR under arbitrary dependence [7].

Algorithm 1: Frequency Noised Mechanism

Input: Grid matrices $\{X^{(t)}\}_{t=1}^T \in \mathbb{R}^{H \times W}$, privacy

parameters (ϵ, δ) , FDR level q , DCT matrices C_x, C_y

Output: Full-size truncated noisy spectra

$\{\Theta_{\text{noisy}}^{(t), \text{trunc}}\}_{t=1}^T \in \mathbb{R}^{H \times W}$, where coefficients with $u > U^*$ or $v > V^*$ are set to 0

```

1 for  $t \leftarrow 1$  to  $T$  do
2    $\Theta_{\text{true}}^{(t)} \leftarrow C_x^T X^{(t)} C_y$ ;
3    $\sigma \leftarrow \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}$ ;
4   for  $t \leftarrow 1$  to  $T$  do
5     Sample  $Z^{(t)} \sim \mathcal{N}(0, \sigma^2)^{H \times W}$ ;
6      $\Theta_{\text{noisy}}^{(t)} \leftarrow \Theta_{\text{true}}^{(t)} + Z^{(t)}$ ;
7   for  $u \leftarrow 1$  to  $H$  do
8     for  $v \leftarrow 1$  to  $W$  do
9        $S(u, v) \leftarrow \sum_{t=1}^T (\Theta_{\text{noisy}}^{(t)}[u, v])^2 / \sigma^2$ ;
10       $p(u, v) \leftarrow 1 - F_{\chi_T^2}(S(u, v))$ ; // Chi-squared CDF
11 Set  $m \leftarrow HW$  and sort  $\{p(u, v)\}$  as  $p_{(1)} \leq \dots \leq p_{(m)}$ ;
12 Find  $k^* \leftarrow \max\{k \mid p_{(k)} \leq (k/m) q\}$ 
13  $\mathcal{S} \leftarrow \{(u, v) \mid p(u, v) \leq p_{(k^*)}\}$ 
14  $U^* \leftarrow \max_{(u,v) \in \mathcal{S}} u$ ;  $V^* \leftarrow \max_{(u,v) \in \mathcal{S}} v$ ;
15 for  $t \leftarrow 1$  to  $T$  do
16    $\Theta_{\text{noisy}}^{(t), \text{trunc}} \leftarrow 0_{H \times W}$ ;
17    $\Theta_{\text{noisy}}^{(t), \text{trunc}}[1 : U^*, 1 : V^*] \leftarrow \Theta_{\text{noisy}}^{(t)}[1 : U^*, 1 : V^*]$ ;
18 return  $\{\Theta_{\text{noisy}}^{(t), \text{trunc}}\}_{t=1}^T$ ;

```

Algorithm 1 implements the discrete cosine transform and BH-FDR truncation for differentially private DCT coefficients. It first computes the true DCT coefficients for each frame (lines 1–2), then calculates the noise scale σ based on the Gaussian mechanism (line 3), and adds i.i.d. Gaussian noise to generate noisy coefficients (lines 4–6). Then, it computes energy statistics $S(u, v)$ and p-values for each frequency coordinate (lines 7–10). The p-values are sorted, and the BH procedure is applied to find the significant set $\mathcal{S}$ (lines 11–13). Bandwidths U^* and V^* are determined by rectangularizing $\mathcal{S}$ (lines 14–15). Finally we update the $\Theta_{\text{noisy}}^{(t), \text{trunc}}$ by the Bandwidths and the noisy coefficients $\Theta_{\text{noisy}}^{(t)}$. It returns the truncated noisy coefficients for each time slices (lines 19).

6 UNSUPERVISED DENOISING

Differential privacy mandates noise injection that compromises data utility. While self-supervised methods like Recorrupted-to-Recorrupted (R2R) learning offer a theoretical solution in the absence of clean data, they suffer practically from high estimator variance and training instability. In this section, we propose a rigorous framework to resolve these limitations. We first derive an optimal **Control Variate (CV)** estimator to minimize R2R variance, extend it to **Weighted GLS** for heteroscedastic noise, and introduce a **Variance-Aware Polyak Step Size** to ensure robust convergence.

6.1 Variance Reduction via Control Variates

Standard R2R training is impeded by stochastic instability, where random noise resampling induces variance that obscures the true gradient signal. However, unlike black-box settings, the privacy noise mechanism here is explicitly known. We exploit this structure to construct a *control variate*—an zero-mean analytic term strongly correlated with the stochastic loss. This allows us to explicitly subtract noise-induced variance without introducing bias.

We formalize this under the DCT noise model. Let $n \sim \mathcal{N}(0, \Sigma)$ be the privacy noise with factorization $\Sigma = LL^\top$, where $L = \sigma(C_y^{(:,0:V)} \otimes C_x^{(:,0:U)}) \in \mathbb{R}^{d \times K}$. Sample auxiliary noise $z \sim \mathcal{N}(0, I_K)$ and define recorruption terms via $\alpha > 0$:

$$A = L\alpha, \quad B = L\alpha^{-1} \quad (4)$$

The recorrupted pairs are then constructed as follows.

$$y^+ = y + Az = x + n + L\alpha z, \quad y^- = y - Bz = x + n - L\alpha^{-1}z \quad (5)$$

The R2R loss is:

$$L(\theta) = \|F_\theta(y^+) - y^-\|_2^2$$

Using the pair difference $\Delta = y^+ - y^-$, define the quadratic control variate:

$$\Delta = y^+ - y^- = L\gamma z, \quad \text{where } \gamma = \alpha + \alpha^{-1}$$

$$C = \|\Delta\|_2^2 = \gamma^2 z^\top (L^\top L) z$$

Since $L(\theta)$ and C share the same noise realization, C is strongly correlated with $L(\theta)$, yielding the CV estimator

$$\widehat{\mathcal{L}}_{cv}(\theta) = L(\theta) - \lambda(C - \mathbb{E}[C])$$

where $\mathbb{E}[C] = \gamma^2 \text{tr}(\Sigma)$ by Gaussian quadratic-form identities.

THEOREM 3 (UNBIASEDNESS AND OPTIMAL VARIANCE REDUCTION IN CV-R2R). *The estimator $\widehat{\mathcal{L}}_{cv}(\theta)$ satisfies $\mathbb{E}[\widehat{\mathcal{L}}_{cv}(\theta)] = \mathbb{E}[L(\theta)]$, thus preserving R2R's statistical equivalence to the supervised objective. The optimal coefficient minimizing $\text{Var}(\widehat{\mathcal{L}}_{cv})$ is:*

$$\lambda^* = \frac{\text{Cov}(L, C)}{\text{Var}(C)},$$

which achieves the minimal variance:

$$\text{Var}(\widehat{\mathcal{L}}_{cv}) = \text{Var}(L)(1 - \rho_{L,C}^2), \quad \rho_{L,C} = \frac{\text{Cov}(L, C)}{\sqrt{\text{Var}(L) \text{Var}(C)}} \quad (6)$$

with explicit form $\text{Var}(C) = 2\gamma^4 \text{tr}(\Sigma^2)$.

PROOF. We establish the properties of the CV-R2R estimator through a sequence of steps.

Step 1: Zero Expectation of Control Variate Term

The fundamental property enabling control variates is that the centered control variate has zero expectation:

$$\mathbb{E}[C - \mathbb{E}[C]] = \mathbb{E}[C] - \mathbb{E}[C] = 0$$

This follows directly from the linearity of expectation.

Step 2: Unbiasedness of CV-R2R Estimator

Using the zero-expectation property, we establish unbiasedness:

$$\mathbb{E}[\widehat{\mathcal{L}}_{cv}(\theta)] = \mathbb{E}[L(\theta)] - \lambda \cdot \mathbb{E}[C - \mathbb{E}[C]] = \mathbb{E}[L(\theta)] - \lambda \cdot 0 = \mathbb{E}[L(\theta)]$$

This preserves the statistical equivalence from the original R2R framework: $\mathbb{E}[L(\theta)] = \mathbb{E}\|F_\theta(y^+) - x\|_2^2 + \mathbb{E}\|x - y^-\|_2^2$, where the second term does not depend on θ (hence it contributes zero gradient with respect to θ).

Step 3: Variance Analysis and Optimal Coefficient

The variance of the control variate estimator is:

$$\begin{aligned} \text{Var}(\widehat{\mathcal{L}}_{cv}) &= \text{Var}(L - \lambda(C - \mathbb{E}[C])) \\ &= \text{Var}(L) + \lambda^2 \text{Var}(C - \mathbb{E}[C]) - 2\lambda \text{Cov}(L, C - \mathbb{E}[C]) \\ &= \text{Var}(L) + \lambda^2 \text{Var}(C) - 2\lambda \text{Cov}(L, C), \end{aligned} \quad (7)$$

where we use $\text{Var}(C - \mathbb{E}[C]) = \text{Var}(C)$ and $\text{Cov}(L, C - \mathbb{E}[C]) = \text{Cov}(L, C)$ since covariance is shift-invariant.

Differentiating with respect to λ :

$$\frac{\partial}{\partial \lambda} \text{Var}(\widehat{\mathcal{L}}_{cv}) = 2\lambda \text{Var}(C) - 2 \text{Cov}(L, C)$$

Setting to zero yields the optimal coefficient:

$$\lambda^* = \frac{\text{Cov}(L, C)}{\text{Var}(C)}$$

Substituting back gives the minimal variance:

$$\begin{aligned} \text{Var}(\widehat{\mathcal{L}}_{cv}) &= \text{Var}(L) + \left(\frac{\text{Cov}(L, C)}{\text{Var}(C)} \right)^2 \text{Var}(C) - 2 \left(\frac{\text{Cov}(L, C)}{\text{Var}(C)} \right) \text{Cov}(L, C) \\ &= \text{Var}(L) - \frac{\text{Cov}^2(L, C)}{\text{Var}(C)} \\ &= \text{Var}(L) \left(1 - \frac{\text{Cov}^2(L, C)}{\text{Var}(L) \text{Var}(C)} \right) \\ &= \text{Var}(L)(1 - \rho_{L,C}^2) \end{aligned} \quad (8)$$

Step 4: Explicit Variance Calculation

For $C = \gamma^2 z^\top S z$ with $S = L^\top L \succeq 0$, the Gaussian quadratic form variance gives:

$$\text{Var}(z^\top S z) = 2 \text{tr}(S^2)$$

This result follows from the moment generating function of Gaussian quadratic forms. Therefore:

$$\text{Var}(C) = \gamma^4 \cdot 2 \text{tr}(S^2) = 2\gamma^4 \text{tr}((L^\top L)^2) = 2\gamma^4 \text{tr}(L^\top L L^\top L) = 2\gamma^4 \text{tr}(\Sigma^2)$$

This completes the proof. $\square$

COROLLARY 4 (OPTIMAL SCALAR SCALING PARAMETER). *In the scalar case $\alpha \in \mathbb{R}_{>0}$, the factor $\gamma^4 = (\alpha + \alpha^{-1})^4$ in $\text{Var}(C)$ achieves its global minimum at $\alpha = 1$.*

PROOF. Consider $f(\alpha) = (\alpha + \alpha^{-1})^4$ for $\alpha > 0$ and define $g(\alpha) = \alpha + \alpha^{-1}$. Then

$$g'(\alpha) = 1 - \alpha^{-2} = \frac{\alpha^2 - 1}{\alpha^2},$$

so $g'(\alpha) = 0$ iff $\alpha = 1$. Moreover,

$$g''(\alpha) = 2\alpha^{-3} > 0 \quad (\alpha > 0).$$

Hence $\alpha = 1$ is the unique minimizer of $g(\alpha)$ and $g(1) = 2$. Since $f(\alpha) = g(\alpha)^4$ is increasing in $g(\alpha)$ for $g(\alpha) > 0$, it also attains its minimum at $\alpha = 1$, with $f(1) = 16$. $\square$

6.2 Generalization to Weighted Objectives

The truncation in the DCT domain introduces a critical characteristic: the noise variance becomes heterogeneous across the spatial grid. This is because each grid cell's effective noise variance is determined by its corresponding coefficient in the truncated DCT basis. To address this inherent variance heterogeneity, we utilize CV-R2R to a weighted generalized least squares (GLS) objective. Introducing a fixed symmetric positive semidefinite weight matrix $W \succeq 0$ (typically $W = \text{Diag}(1/\text{diag}(\Sigma))$), we define:

$$L_W(\theta) = \|W^{1/2}(F_\theta(y^+) - y^-)\|_2^2, \quad C_W = \Delta^\top W \Delta = \gamma^2 z^\top (L^\top W L) z \quad (9)$$

The weighted CV-enhanced estimator becomes:

$$\widehat{\mathcal{L}}_{\text{cv},W}(\theta) = L_W(\theta) - \lambda(C_W - \mathbb{E}[C_W])$$

with $\mathbb{E}[C_W] = \gamma^2 \text{tr}(W\Sigma)$.

THEOREM 5 (EQUIVALENCE IN GLS-CV-R2R). *Under the weighted setup:*

$$\mathbb{E}[\widehat{\mathcal{L}}_{\text{cv},W}(\theta)] = \mathbb{E}\|W^{1/2}(F_\theta(y^+) - x)\|_2^2 + \mathbb{E}\|W^{1/2}(x - y^-)\|_2^2,$$

where the second term does not depend on θ (and thus contributes zero gradient with respect to θ). Thus minimizing $\widehat{\mathcal{L}}_{\text{cv},W}$ is equivalent to optimizing the GLS-supervised objective, with variance reduction governed by Theorem 1.

PROOF. From the weighted R2R equivalence theorem:

$$\mathbb{E}[L_W(\theta)] = \mathbb{E}\|W^{1/2}(F_\theta(y^+) - x)\|_2^2 + \mathbb{E}\|W^{1/2}(x - y^-)\|_2^2$$

The constant term expands explicitly as:

$$\mathbb{E}\|W^{1/2}(x - y^-)\|_2^2 = \mathbb{E}[(n - Bz)^\top W(n - Bz)] = \text{tr}(W\Sigma) + \text{tr}(WBB^\top)$$

because $n \perp z$ ensures vanishing cross terms.

To verify the conditional expectation structure, define $u = n + Az$ and $v = n - Bz$. Since $A = L\alpha$ and $B = L\alpha^{-1}$ with scalar $\alpha > 0$, we have

$$AB^\top = L\alpha(\alpha^{-1}L^\top) = LL^\top = \Sigma,$$

so

$$\text{Cov}(u, v) = \Sigma - AB^\top = LL^\top - LL^\top = 0.$$

Because (u, v) are jointly Gaussian, zero covariance implies independence: $u \perp v$.

Decomposing the weighted loss conditionally:

$$\begin{aligned} & \mathbb{E}[\|W^{1/2}(F_\theta(y^+) - y^-)\|_2^2 | y^+, x] \\ &= \mathbb{E}[\|W^{1/2}(F_\theta(y^+) - x + x - y^-)\|_2^2 | y^+, x] \\ &= \|W^{1/2}(F_\theta(y^+) - x)\|_2^2 + \mathbb{E}[\|W^{1/2}(x - y^-)\|_2^2 | y^+, x] \\ &\quad + 2\langle W^{1/2}(F_\theta(y^+) - x), W^{1/2}\mathbb{E}[x - y^- | y^+, x] \rangle \\ &= \|W^{1/2}(F_\theta(y^+) - x)\|_2^2 + \mathbb{E}[\|W^{1/2}(x - y^-)\|_2^2 | y^+, x] \end{aligned} \quad (10)$$

as the cross term vanishes. Taking full expectation confirms the decomposition.

Unbiasedness of the CV term follows similarly to Theorem 1, and variance optimization proceeds identically with weighted quantities, yielding $\text{Var}(C_W) = 2\gamma^4 \text{tr}((W\Sigma)^2)$. $\square$

6.3 Variance-Aware Polyak Step Size

While CV-R2R reduces the loss-estimation variance, optimization remains challenging under heterogeneous privacy noise and evolving signal-to-noise ratios. R2R's dual-noise sampling provides two independent gradient realizations, enabling not only an unbiased gradient estimate but also a direct proxy for gradient-noise intensity, which we leverage to design a variance-aware Polyak-type step-size rule.

6.3.1 Dual Recorruption Architecture and Variance Estimation. The mathematical foundation of CV-SPS rests upon exploiting the inherent parallelism within the R2R framework. At each training iteration, we sample two independent noise vectors from the established Gaussian distribution:

$$z^{(a)}, z^{(b)} \sim \mathcal{N}(0, I_K), \quad z^{(a)} \perp z^{(b)}$$

These independent samples generate dual recorrupted pairs through the previously defined transformations:

$$y^{+(i)} = y + Az^{(i)}, \quad y^{-(i)} = y - Bz^{(i)}, \quad i \in \{a, b\}$$

where $A = L\alpha$ and $B = L\alpha^{-1}$ maintain the statistical equivalence properties established in the CV-R2R derivation. From these pairs, we compute two independent realizations of the loss function and their corresponding gradients:

$$L^{(i)}(\theta) = \|F_\theta(y^{+(i)}) - y^{-(i)}\|_2^2, \quad g^{(i)} = \nabla_\theta L^{(i)}(\theta), \quad i \in \{a, b\}$$

The critical observation is that these dual computations provide the necessary statistics for simultaneous gradient estimation and variance quantification.

THEOREM 6 (UNBIASED GRADIENT AND VARIANCE ESTIMATION). *For gradients $g^{(a)}, g^{(b)}$ computed from independent noise realizations conditioned on parameters θ , define the averaged gradient $\bar{g} = \frac{1}{2}(g^{(a)} + g^{(b)})$ and variance proxy $v = \frac{1}{4}\|g^{(a)} - g^{(b)}\|_2^2$. Then the following equalities hold:*

$$\mathbb{E}[\bar{g}] = \mathbb{E}[g], \quad \mathbb{E}[v] = \frac{1}{2}\mathbb{E}[\|g - \mathbb{E}[g]\|_2^2]$$

with the expectations taken over the noise distribution.

PROOF. The unbiasedness of the gradient estimate follows directly from the linearity of expectation:

$$\mathbb{E}[\bar{g}] = \frac{1}{2}(\mathbb{E}[g^{(a)}] + \mathbb{E}[g^{(b)}]) = \mathbb{E}[g]$$

For the variance proxy, we analyze the expectation of the squared gradient difference:

$$\begin{aligned}\mathbb{E}[\|g^{(a)} - g^{(b)}\|^2] &= \mathbb{E}[\|g^{(a)}\|^2] + \mathbb{E}[\|g^{(b)}\|^2] - 2\mathbb{E}[\langle g^{(a)}, g^{(b)} \rangle] \\ &= 2\mathbb{E}[\|g\|^2] - 2\|\mathbb{E}[g]\|^2\end{aligned}\quad (11)$$

The simplification of the cross-term relies on the independence of $g^{(a)}$ and $g^{(b)}$. More specifically, in each iteration, θ is fixed, and $z^{(a)}$ and $z^{(b)}$ are independent; therefore, the resulting gradients $g^{(a)}$ and $g^{(b)}$ generated from them are independent (conditional on the current sample/mini-batch).

$$\begin{aligned}\mathbb{E}[\langle g^{(a)}, g^{(b)} \rangle \mid \theta] &= \mathbb{E}\left[\left(g^{(a)}\right)^\top g^{(b)} \mid \theta\right] \\ &= \left(\mathbb{E}[g^{(a)} \mid \theta]\right)^\top \left(\mathbb{E}[g^{(b)} \mid \theta]\right) \quad (\text{since } g^{(a)} \perp g^{(b)} \mid \theta) \\ &= \langle \mathbb{E}[g^{(a)} \mid \theta], \mathbb{E}[g^{(b)} \mid \theta] \rangle \\ &= \|\mathbb{E}[g \mid \theta]\|_2^2.\end{aligned}$$

Therefore, the expectation of the variance proxy becomes:

$$\mathbb{E}[v] = \frac{1}{4} \cdot 2(\mathbb{E}[\|g\|^2] - \|\mathbb{E}[g]\|^2) = \frac{1}{2} \text{Var}(g) \quad (12)$$

This establishes that v serves as an unbiased estimator of gradient noise intensity, scaled by a constant factor of one-half. $\square$

6.3.2 Stochastic Polyak Step-size with Variance Adaptation. The classical Polyak step size provides theoretical optimality for deterministic optimization but proves inadequate in stochastic settings due to unaccounted gradient noise. We derive a stochastic variant that explicitly incorporates variance information through the following formulation:

$$\eta_t = \text{clip}\left(\frac{\hat{L}(\theta_t) - L_\star}{\|\bar{g}_t\|^2 + \beta v_t + \varepsilon}, \eta_{\min}, \eta_{\max}\right),$$

where $\hat{L}(\theta_t) = \frac{1}{2}(L^{(a)}(\theta_t) + L^{(b)}(\theta_t))$ represents an unbiased loss estimate, $\bar{g}_t$ denotes the unbiased gradient estimate, v_t quantifies gradient noise intensity, $L_\star$ establishes a lower bound on achievable loss, β modulates conservatism in low signal-to-noise regimes, and ε ensures numerical stability.

The theoretical justification for $L_\star$ stems from the control variate correction within CV-R2R. Since the control variate mechanism removes noise bias, setting $L_\star = 0$ becomes theoretically justified, though a small negative buffer $L_\star = -\gamma\sigma^2$ may accommodate residual uncertainty in practical implementations.

PROPOSITION 1 (EXPECTED DESCENT OF IDEALIZED CV-SPS). Assume $J(\theta)$ is L -smooth and satisfies the Polyak-Łojasiewicz (PL) condition

$$\frac{1}{2}\|\nabla J(\theta)\|_2^2 \geq \mu(J(\theta) - J_\star), \quad \mu > 0.$$

Let the stochastic gradient satisfy

$$\mathbb{E}[g_t \mid \theta_t] = \nabla J(\theta_t), \quad \mathbb{E}\|g_t - \nabla J(\theta_t)\|_2^2 \leq \sigma^2.$$

Consider the idealized step size

$$\eta_t = \frac{J(\theta_t) - J_\star}{\|\nabla J(\theta_t)\|_2^2 + \beta\sigma^2}, \quad 0 < \beta \leq \frac{1}{2L},$$

Table 2: Statistics of datasets

Datasets	#Type	#Area	#Scale
4SQ_TKY	Real	Tokyo	755,885
CABS_SF	Real	San Francisco	568,000
Gowalla_SF	Real	San Francisco	846,000

and the update $\theta_{t+1} = \theta_t - \eta_t g_t$. Then there exists a constant $c \in (0, 1)$ depending only on $(L, \mu, \beta, \sigma^2)$ such that

$$\mathbb{E}[J(\theta_{t+1}) - J_\star] \leq (1 - c) \mathbb{E}[J(\theta_t) - J_\star].$$

In particular, the idealized CV-SPS scheme converges linearly in expectation.

SKETCH. By L -smoothness,

$$J(\theta_{t+1}) \leq J(\theta_t) + \langle \nabla J(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{L}{2} \|\theta_{t+1} - \theta_t\|_2^2.$$

Substituting $\theta_{t+1} - \theta_t = -\eta_t g_t$ and taking the conditional expectation given the following θ_t , using $\mathbb{E}[g_t \mid \theta_t] = \nabla J(\theta_t)$ and $\mathbb{E}\|g_t\|_2^2 \leq \|\nabla J(\theta_t)\|_2^2 + \sigma^2$, yields an upper bound:

$$\mathbb{E}[J(\theta_{t+1}) \mid \theta_t] \leq J(\theta_t) - \eta_t \|\nabla J(\theta_t)\|_2^2 + \frac{L}{2} \eta_t^2 (\|\nabla J(\theta_t)\|_2^2 + \sigma^2).$$

Substituting in the definition of η_t and simplifying, one can show that for $0 < \beta \leq 1/(2L)$ the right-hand side is bounded by

$$J(\theta_t) - c_0(J(\theta_t) - J_\star)$$

for some constant $c_0 > 0$ that depends on $(L, \mu, \beta, \sigma^2)$ via the PL inequality $\|\nabla J(\theta_t)\|_2^2 \geq 2\mu(J(\theta_t) - J_\star)$. Taking total expectation over θ_t gives the claimed contraction with $c = c_0$. $\square$

Practical implementation. In practice, we replace $J(\theta_t)$ and $\nabla J(\theta_t)$ by their unbiased estimators $\hat{L}(\theta_t)$ and $\bar{g}_t$, and replace the unknown variance σ^2 by the proxy v_t derived in Theorem 6. We also apply clipping and add a small $\varepsilon > 0$ for numerical stability, which can be interpreted as a robustified approximation of the above idealized Polyak-type rule.

7 EXPERIMENTS

We evaluate PrivSTD against seven baselines across cell count, hotspot, and forecasting tasks. Due to space constraints, the **Ab-lation Study** and the **User-Level extension** are provided in the Appendix of the full version [5].

7.1 Experimental Setup

Datasets. We use three public real-world spatiotemporal datasets: (1) **4SQ_TKY** [50], the Tokyo subset of Foursquare dataset containing 755,885 location updates from 8,000 users over 22 months; (2) **Gowalla_SF** [13], the San Francisco subset of Gowalla dataset with 568,000 location updates from 14,000 users over 575 days; (3) **CABS_SF** [38], taxi GPS trajectories in San Francisco with 846,000 records over 575 days. Following the VDR framework [4], we partition the spatial domain into grids of varying resolutions (128×128 to 768×768) to achieve different spatial granularities. The temporal domain is discretized into 24 slices (monthly aggregation). **Experimental Settings.** We study the impact of the parameters listed in Table 3. Our method is implemented in Python 3.90, and the experiments are conducted on a computer with an Intel Xeon

Table 3: Parameters table

Parameter	Range
the privacy budget ϵ	0.1, 0.5, 1, 2, 3, 4, 5
the grid number H	128, 256, 384 , 512, 640, 768

2.1GHz CPU, 128 GB main memory, and a RTX3090 GPU. The source code is publicly available [5].

Evaluation Metrics. We evaluate the released spatiotemporal density tensor using three representative query tasks: **cell count**, **hotspot search**, and **temporal forecasting**.

Cell Count. For randomly sampled cells (i, j, t) , we measure the relative error $RE(y, u) = \frac{|y-u|}{\max\{u, \psi\}}$, where $u = H[i, j, t]$, $y = \hat{H}[i, j, t]$, and ψ stabilizes small counts.

Hotspot. Given a query location q , a threshold ν , and a bounded region SR , we return the nearest cell in SR with density $> \nu$ (or the max-density cell if none exists), and we report MAE between the true and reported hotspot distances.

Forecasting. We forecast each cell’s future count with horizon $h = 3$ under a holdout protocol and evaluate the result using sMAPE. With monthly aggregation ($T = 24$), we set seasonal period $sp = 3$, keeping only series with length $\geq 3sp$ and significant autocorrelation at lag sp (90% confidence). Results are averaged over a large set of randomized queries.

Baseline Methods. We compare with against seven state-of-the-art differentially private methods: (1) **Uniform Grid (UG)** [40] with equal-sized spatial partitions; (2) **Adaptive Grid (AG)** [40] with data-dependent spatial partitioning; (3) **HB_stripped** [41] applying hierarchical histograms per time slice; (4) **PrivBayes** [53] using Bayesian network synthesis; (5) **AHP** [54] with adaptive histogram publication; (6) **MWEM** [22] employing the multiplicative weights exponential mechanism; (7) **VDR** [4] employing VQ-VAE as the denoising model for post-processing. Here, VDR^1 denotes the variant without multi-resolution learning, while VDR^2 denotes the variant with multi-resolution learning. To ensure a fair comparisons, all methods employ the Gaussian mechanism to achieve approximate differential privacy, utilizing the optimal parameter configurations as reported in their original publications.

7.2 Overall Findings

Our comparison of PrivSTD with the seven state-of-the-art baseline methods provides evidence that PrivSTD is capable of the highest utility across all query types. This performance is due to PrivSTD’s fundamental reconstruction of the noise injection paradigm through two synergistic innovations: (i) the Discrete Cosine Transform (DCT) mechanism that perturbs data in a structurally-aligned frequency domain (ii) and the Control Variate Enhanced Recorrupted-to-Recorrupted (CV-R2R) network that enables high-fidelity unsupervised denoising.

Result Visualization. To provide visual insight into the advantages of PrivSTD, we visualize density data from the CABS_SF dataset as heatmaps in Figure 4. The heatmaps illustrate density distributions across grid cells, where brighter colors indicate higher densities. The visualization shows that PrivSTD and VDR generate

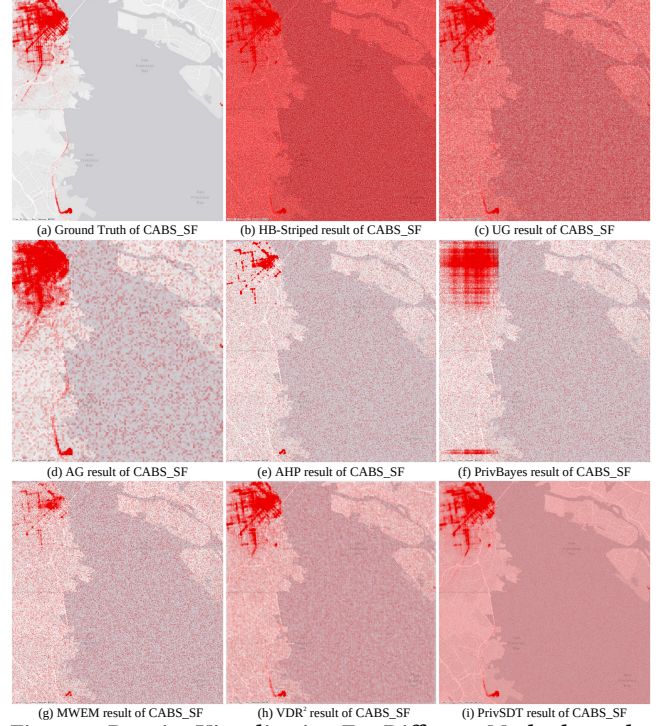


Figure 4: Density Visualization For Different Methods on the CABS_SF Dataset with $\epsilon = 0.5$.

trajectories with superior fidelity to the ground truth. Next, Table 4 provides detailed performance comparisons across the three practical query types discussed previously.

Results in terms of Cell count. PrivSTD yields the lowest relative error across all datasets, particularly under strong privacy guarantees. The conventional grid-based methods (UG, AG, HB-Striped) operate independently on each temporal slice and inject noise directly in to the spatial domain, which disrupts spatiotemporal consistency and amplifies errors in high-dimensional settings. Synthesis-based methods such as PrivBayes struggle to model complex joint distributions over space and time, while the learning-based VDR remains limited by its spatial-domain noise injection, where structural information is already degraded markedly before denoising. In contrast, PrivSTD perturbs data in the frequency domain and preserves dominant low-frequency structures, enabling the CV-R2R network to reconstruct accurate densities from a compact and structure-aware representation.

Results in terms of Hotspot Query. PrivSTD outperforms all baselines at hotspot localization, achieving the smallest mean absolute error (MAE). The spatial-domain perturbation in UG, AG, AHP, and MWEM tends to blur sharp density gradients and hotspot boundaries, especially in heterogeneous regions with mixed dense and sparse sub-regions. Although VDR improves upon the other methods, its denoising objective focuses on recovering noisy spatial grid cells rather than explicitly on preserving local extrema. In contrast, PrivSTD combines frequency-domain noise suppression with variance-reduced reconstruction, allowing it to retain sharp local density contrasts and more accurately recover hotspot locations.

Results in terms of Forecasting Query. PrivSTD also shows a clear advantage when considering temporal forecasting. Most

Table 4: Overall Utility Performance with Different Privacy Budgets (best values are in bold; smaller values are better)

Metric	Method	4SQ_TKY				CABS_SF				Gowalla_SF			
		$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$
Cell Count	UG	1.279	0.487	0.164	0.137	1.211	0.469	0.364	0.335	1.581	0.649	0.235	0.227
	AG	0.348	0.342	0.310	0.292	0.899	0.698	0.501	0.454	0.471	0.421	0.373	0.335
	HB-Striped	1.461	0.731	0.365	0.243	1.510	0.755	0.335	0.251	1.565	0.782	0.391	0.261
	AHP	0.407	0.363	0.292	0.248	0.513	0.507	0.446	0.369	0.409	0.404	0.373	0.321
	MWEM	0.385	0.369	0.342	0.314	0.473	0.454	0.420	0.390	0.382	0.369	0.342	0.319
	VDR ¹	0.262	0.165	0.124	0.116	0.341	0.229	0.174	0.171	0.287	0.179	0.145	0.125
	VDR ²	0.237	0.148	0.110	0.093	0.299	0.185	0.174	0.119	0.239	0.167	0.121	0.106
	PrivBayes	0.351	0.349	0.310	0.304	0.349	0.344	0.333	0.333	0.282	0.275	0.270	0.256
	PrivSTD ¹	0.149	0.119	0.073	0.057	0.280	0.154	0.099	0.075	0.194	0.128	0.086	0.067
Hotspot Query	UG	10.023	8.092	1.851	1.296	5.267	4.191	3.897	3.172	6.238	5.732	4.970	2.397
	AG	5.454	5.172	4.703	4.921	4.368	2.574	2.147	2.238	4.809	4.596	3.858	3.172
	HB-Striped	10.378	10.203	9.497	7.727	3.968	3.805	3.458	2.753	5.765	5.588	5.054	3.724
	AHP	25.223	11.142	4.578	1.841	42.258	14.120	2.817	0.921	95.914	23.819	6.014	0.811
	MWEM	21.553	19.611	12.114	11.207	28.083	24.789	17.733	10.599	30.332	26.676	26.366	23.702
	VDR ¹	1.611	0.841	0.948	0.925	4.126	2.007	1.952	1.972	2.397	1.683	1.712	1.486
	VDR ²	1.646	0.930	0.738	0.673	2.311	1.926	1.803	1.776	1.504	1.585	1.453	1.426
	PrivBayes	16.247	13.324	11.420	10.736	13.433	12.687	12.058	10.839	21.146	19.640	17.893	17.599
	PrivSTD ¹	1.364	0.415	0.298	0.127	3.397	1.216	0.491	0.303	2.261	0.652	0.315	0.304
Forecasting Query	UG	1.551	1.421	1.097	1.073	1.091	0.948	0.829	0.651	1.586	1.351	1.171	1.002
	AG	1.217	1.169	1.150	1.132	0.977	0.789	0.795	0.703	1.228	1.148	1.126	1.092
	HB-Striped	1.731	1.584	1.374	1.284	1.018	0.686	0.533	0.484	1.691	1.536	1.328	1.196
	AHP	1.297	1.259	1.171	1.119	1.607	1.599	1.433	1.196	1.287	1.072	1.069	1.051
	MWEM	1.295	1.236	1.189	1.088	1.379	1.302	1.133	1.027	1.245	1.140	1.004	0.976
	VDR ¹	1.794	1.747	1.745	1.741	1.803	1.793	1.786	1.785	1.926	1.917	1.913	1.916
	VDR ²	1.801	1.770	1.747	1.737	1.795	1.789	1.783	1.782	1.930	1.924	1.920	1.915
	PrivBayes	1.293	1.218	1.153	1.069	0.873	0.849	0.854	0.859	1.214	1.053	1.015	1.006
	PrivSTD ¹	1.175	1.159	1.068	1.025	0.657	0.458	0.435	0.421	1.199	1.025	0.943	0.925

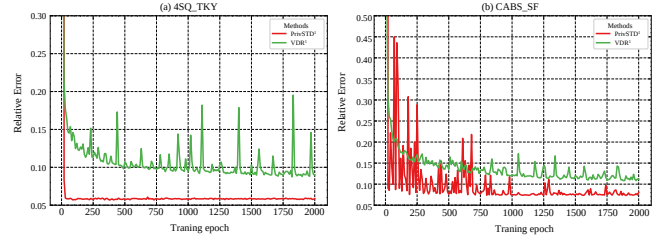
baselines lack explicit temporal modeling and treat each time slice core independently, while PrivBayes captures only limited attribute correlations. VDR performs poorly due to its purely spatial architecture, which cannot exploit temporal dependencies. PrivSTD benefits from the sinusoidal basis of DCT, which naturally captures periodic patterns, and from its temporally consistent coefficient reconstruction, leading to lower sMAPE across all datasets.

7.3 Convergence and Runtime

Figure 5 compares the training convergence of PrivSTD¹ and VDR. PrivSTD¹ consistently converges faster and stabilizes at a lower relative error across datasets, indicating more effective suppression of DP noise during training. In contrast, VDR exhibits slower convergence and larger oscillations, reflecting higher variations in its denoising objective.

We also observe dataset-dependent convergence behavior. On 4SQ_TKY, which has a shorter temporal span, PrivSTD¹ rapidly reaches a stable error plateau, whereas on CABS_SF with its longer time series, early-stage fluctuations are more pronounced. Occasional large initial RE values arise from truncation-induced heteroscedastic reconstructions and sparsity-sensitive errors, which gradually diminish as the training stabilizes.

Table 5 summarizes the end-to-end runtime, where **Infer** denotes the cost of publishing a sanitized dataset once. Traditional non-learning mechanisms incur no training cost but require a relatively expensive publishing procedure each time. Learning-based methods

**Figure 5: Relative Error over Training Epochs****Table 5: Training and Inference Time Across Datasets**

Method	Dataset					
	4SQ_TKY		CABS_SF		Gowalla_SF	
	Train	Infer	Train	Infer	Train	Infer
UG	-	0.252	-	0.286	-	0.249
AG	-	3.502	-	3.089	-	9.03
HB-Striped	-	14.364	-	15.164	-	17.547
AHP	-	5.906	-	5.76	-	6.211
MWEM	-	13.776	-	4.818	-	10.906
PrivBayes	-	126.69	-	131.953	-	93.748
VDR ²	1206.59	0.033	1192.57	0.036	1215.743	0.049
PrivSTD ¹	409.31	0.043	545.76	0.047	434.192	0.043

Training and inference times are in seconds.

introduce a one-time offline training overhead, after which each release is generated by a single forward pass. Compared to VDR, PrivSTD¹ reduces the training time considerably while maintaining

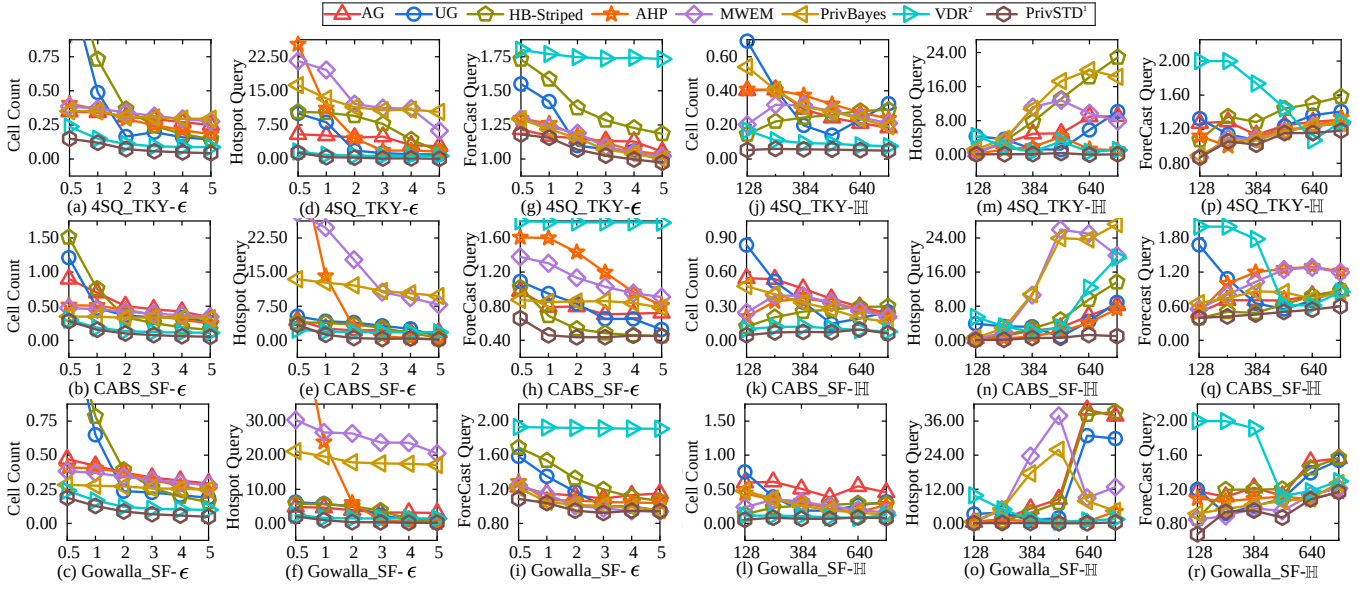


Figure 6: Parameter Sensitivity

a similarly low inference cost, making it more suitable for repeated data releases.

7.4 Parameter Sensitivity Analysis

The default parameter settings employed in Figures 6 are shown in bold in Table 3 (set as the default values). When investigating the sensitivity of any individual parameter, all other parameters are held constant to ensure controlled experimental conditions.

Impact of parameter ϵ . As the privacy budget ϵ increases, all methods exhibit decreasing errors across the three query types, consistent with the fundamental privacy-utility trade-off of differential privacy. Throughout this range, PrivSTD consistently maintains the best performance, benefiting from its frequency-domain noise injection and variance-reduced unsupervised denoising. In contrast, VDR shows limited improvement for Cell count and hotspot queries and performs poorly at forecasting, reflecting its restricted ability to capture temporal patterns. Interestingly, AG is surpassed by UG at larger ϵ , indicating that partitioning variance becomes the dominant error source when noise diminishes. Structure-dependent methods such as AHP and HB-Striped are highly sensitive to ϵ : strong noise disrupts their partitioning decisions under small ϵ , while reduced noise at larger ϵ allows partial recovery of intrinsic data structures.

Impact of parameter H . As the grid parameter H increases and the grids becomes finer, the count within each cell decreases, so the fixed variance noise injected by differential privacy is amplified on a relative scale, and the signal to noise ratio degrades. Consequently, most methods incur higher errors for the *Hotspot Query* and the *Forecast Query* because hotspot detection relies on local density contrast that is highly sensitive to fine perturbations, while forecasting requires temporally coherent and learnable patterns that are easily destabilized when noise dominates. Interestingly, the *Cell Count* error can decrease overall, and UG exhibits a decrease-then-increase trend. Real trajectory densities are highly heterogeneous

with clustered hotspots and large sparse regions, so with coarse grids, the within-cell uniformity approximation induces substantial discretization and non uniformity bias that may exceed the DP noise. As a grid is refined, smaller cells and more accurate boundary approximation reduce this bias, and over a broad range of H , the bias reduction can outweigh the growing noise effect. In addition, post processing such as non negativity projection, smoothing, and denoising tend to push empty cell outputs toward zero and to shift the average metric downward, so the reported error in high-resolution settings decreases. The UG curve highlights the trade off features among discretization induced bias, perturbation noise error, and the additional relative bias introduced by post processing.

8 CONCLUSION

We propose PrivSTD, a two-stage method for publishing high-resolution spatio-temporal density data with differential privacy guarantees. PrivSTD first performs private frequency estimation to capture dominant spatial correlations in the frequency domain while providing rigorous event-level privacy guarantees. By incorporating statistically controlled frequency truncation, the method effectively suppresses noise-dominated components and derives an explicit noise covariance structure. Furthermore, PrivSTD leverages covariance-aware self-supervised denoising based on statistically consistent recorruption to reduce structured and heteroscedastic privacy noise without accessing the original data, enabling stable learning under strict privacy constraints. An experimental study on real-world mobility datasets shows that PrivSTD consistently achieves high utility and is capable of outperforming state-of-the-art baselines.

Table 6: Overall Utility Performance of Different Variant(best values are in bold; smaller values are better).

Metric	Method	4SQ_TKY				CABS_SF				Gowalla_SF			
		$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$
Cell count	PrivSTD ¹	0.149	0.119	0.073	0.057	0.280	0.154	0.099	0.075	0.194	0.128	0.086	0.067
	PrivSTD ²	0.152	0.130	0.074	0.056	0.294	0.161	0.104	0.076	0.199	0.126	0.091	0.067
	PrivSTD ³	0.205	0.149	0.075	0.057	0.317	0.164	0.109	0.083	0.210	0.133	0.097	0.068
	PrivSTD ⁴	0.325	0.190	0.141	0.089	0.513	0.507	0.446	0.369	0.300	0.173	0.101	0.097
	OnlyDCT ¹	0.409	0.228	0.123	0.084	0.473	0.454	0.420	0.390	0.296	0.223	0.122	0.085
	OnlyDCT ²	0.326	0.208	0.122	0.083	0.341	0.229	0.174	0.171	0.273	0.213	0.121	0.075
	R2R ¹	0.365	0.278	0.151	0.112	0.399	0.273	0.188	0.136	0.356	0.212	0.163	0.126
	R2R ²	0.237	0.206	0.110	0.091	0.387	0.272	0.167	0.127	0.305	0.185	0.128	0.099
Hotspot Query	PrivSTD ¹	1.364	0.415	0.298	0.127	3.397	1.216	0.491	0.303	2.261	0.652	0.315	0.304
	PrivSTD ²	5.163	1.488	0.508	0.196	4.368	2.574	2.147	2.238	2.855	0.734	0.407	0.374
	PrivSTD ³	5.767	1.674	0.530	0.316	3.968	3.805	3.458	2.753	3.166	0.698	0.447	0.397
	PrivSTD ⁴	4.485	3.722	0.495	0.246	6.806	3.708	0.588	0.387	3.760	1.344	0.965	0.483
	OnlyDCT ¹	9.860	8.104	0.761	0.238	9.352	2.866	0.579	0.325	4.410	2.459	0.542	0.421
	OnlyDCT ²	5.954	2.514	0.543	0.326	4.126	2.007	1.952	1.972	3.643	2.131	0.506	0.409
	R2R ¹	7.943	6.519	1.818	0.339	8.740	6.302	1.629	0.661	4.774	2.764	0.786	0.696
	R2R ²	5.216	4.257	0.584	0.284	7.277	4.512	1.095	0.577	4.170	2.329	1.889	0.630
Forecasting Query	PrivSTD ¹	1.175	1.159	1.068	1.025	0.657	0.458	0.435	0.421	1.199	1.025	0.943	0.925
	PrivSTD ²	1.217	1.184	1.072	1.034	0.671	0.466	0.446	0.444	1.211	1.061	0.965	0.961
	PrivSTD ³	1.316	1.205	1.097	1.046	0.711	0.472	0.447	0.444	1.229	1.058	0.970	0.968
	PrivSTD ⁴	1.321	1.237	1.159	1.097	0.848	0.596	0.455	0.447	1.301	1.152	1.004	1.038
	OnlyDCT ¹	1.385	1.239	1.117	1.067	0.764	0.524	0.463	0.458	1.267	1.146	1.006	0.976
	OnlyDCT ²	1.216	1.180	1.100	1.067	0.726	0.504	0.455	0.468	1.245	1.140	0.994	0.970
	R2R ¹	1.302	1.265	1.172	1.156	0.949	0.667	0.482	0.464	1.456	1.277	1.179	1.005
	R2R ²	1.190	1.168	1.112	1.091	0.847	0.615	0.474	0.452	1.360	1.183	1.081	0.999

9 APPENDIX

9.1 Abalation Study

Table 7 summarizes the ablation variants of **PrivSTD**, where each variant isolates a specific design choice in the proposed framework. In particular, we investigate the impact of the noise injection domain (spatial vs. frequency), the use of DCT truncation, the recovery model, and the optimizer. Variants labeled *PrivSTD* correspond to controlled modifications of the full model, while *OnlyDCT* and *R2R* serve as simplified baselines to disentangle the individual contributions of frequency-domain perturbation and recovery-based optimization.

Table 7: Summary of ablation variants in PrivSTD.

Variant	Noise Domain	BH-FDR	Recovery	Optimizer
PrivSTD ¹	Frequency	✓	CV-R2R	CV-SPS
PrivSTD ²	Frequency	✓	CV-R2R	Adam
PrivSTD ³	Frequency	×	CV-R2R	CV-SPS
PrivSTD ⁴	Spatial	–	CV-R2R	CV-SPS
OnlyDCT ¹	Frequency	×	–	–
OnlyDCT ²	Frequency	✓	–	–
R2R ¹	Spatial	–	R2R	Adam
R2R ²	Frequency	✓	R2R	Adam

Table 6 summarizes the ablation results of different PrivSTD variants. The complete model, PrivSTD¹, combines Truncated DCT, CV-R2R, and CV-SPS, while other variants selectively remove or replace individual components to isolate their effects.

Effect of Noise Injection Domain.. Frequency-domain noise injection consistently outperforms spatial-domain perturbation. Comparing PrivSTD¹ with PrivSTD⁴ shows that injecting independent noise directly into the spatial grid severely disrupts local gradients and hotspot boundaries, increasing the recovery burden. In contrast, DCT-domain perturbation separates dominant low-frequency structures from noise at the source, leading to substantially better recoverability. This advantage is further confirmed by the comparison between R2R² and R2R¹, demonstrating that frequency-domain noise benefits recovery even without PrivSTD-specific designs.

Effect of Truncated DCT.. Within the frequency domain, significance-based truncation plays a critical role under strong privacy. PrivSTD³, which uses Full DCT without truncation, retains many high-frequency coefficients dominated by DP noise, resulting in spatial artifacts after inverse transformation. Truncated DCT effectively removes these noise-dominated components, yielding cleaner inputs for the recovery network. This effect is most pronounced at small ϵ , while the performance gap narrows as noise decreases. The same trend is observed when comparing OnlyDCT² with OnlyDCT¹, indicating that the benefit arises from frequency-domain structure rather than the recovery network.

Effect of Recovery Objective and Optimization.. Finally, the comparisons among PrivSTD¹, R2R², and PrivSTD² highlight the importance of variance reduction in both the denoising objective and optimization. CV-R2R substantially improves stability over standard R2R by reducing estimation variance under the recorrup-tion mechanism. Replacing CV-SPS with Adam (PrivSTD²) yields similar average performance but introduces less stable behavior under strong noise. Although the gain from CV-SPS is moderate,

it provides essential robustness for training under heteroscedastic DP noise.

9.2 Extension to User-level DP

In this section, we extend all baseline methods to the **user-level DP** setting. Unlike *event-level DP*, which protects individual trajectory points and assumes that each event contributes independently to the dataset, *user-level DP* treats all trajectory points generated by the same user as a single unit and aims to protect the presence or absence of an entire user. As a result, user-level DP provides a strictly stronger privacy guarantee but also leads to substantially higher sensitivity. Specifically, under user-level DP, the global sensitivity increases from 1 to $\sqrt{k}$, where k denotes the maximum number of trajectory points contributed by any user. Since k can be very large in practice, directly adding noise based on this sensitivity may cause the signal to be overwhelmed by noise. To mitigate this issue, we adopt a commonly used sampling strategy. We set a smaller sampling threshold k_s : for users with no more than k_s trajectory points, all points are retained; for users with more than k_s points, k_s trajectory points are randomly sampled. Noise is then injected into the spatio-temporal trajectory density computed from this sampled dataset to obtain an initial noisy result.

Due to the use of sampling, the reconstructed data may exhibit a systematic reduction in magnitude. Following the recommendation in [4], we therefore apply a scaling factor $\gamma = |D_s|/|D|$ during the final reconstruction stage, where $|D_s|$ and $|D|$ denote the sizes of the sampled dataset and the original dataset, respectively, to compensate for the loss caused by sampling. It is worth noting that VDR adopts its own sampling and reconstruction mechanism under the user-level DP setting; in our experiments, we use the original user-level implementation of VDR without modification.

Dataset. Table 8 has summarized the information of datasets. We use three public real-world spatiotemporal datasets: (1) **Tokyo** [14], the Tokyo subset of Brightkite dataset containing 133,593 check-in point updates from 2,105 users over 30 months; (2) **Stockholm** [13], the Stockholm subset of Gowalla dataset with 568,000 location updates from 8,780 users over 443 days; (3) **Austin** [13], the Austin subset of Gowalla dataset with 212,514 location updates from 7,099 users over 589 days. We partition the spatial domain into 384×384 grid cells. The temporal domain is discretized into 24 slices. Table 9 show the overall result.

Table 8: Statistics of datasets

Datasets	#Type	#Area	#Scale	#Users	k_s
Tokyo	Real	Tokyo	133,593	2,105	20
Stockholm	Real	Stockholm	141,262	8,780	5
Austin	Real	Austin	212,514	7,099	3

9.2.1 Overall Finding. Results in terms of Cell Count. Under the user-level DP setting, the degradation in Cell Count utility is more pronounced than in the event-level case, mainly due to the higher global sensitivity caused by each user contributing multiple trajectory points. In this stronger perturbation regime, methods that inject noise directly in the spatial domain commonly suffer from structural degradation and early performance saturation. Grid-based approaches such as UG and AG add noise independently to

each cell, which inevitably disrupts spatial and temporal consistency. Hierarchical methods like HB-Striped are particularly fragile under high-dimensional spatio-temporal data, as the privacy budget must be divided across multiple levels. Optimization-based methods, including AHP and MWEM, are further constrained by the combined effects of approximation error and amplified noise. Although the learning-based method VDR generally outperforms traditional baselines, its noise is still injected in the spatial domain, irreversibly damaging structural information before denoising. As a result, even when the privacy budget increases, its performance improvement remains limited and gradually saturates.

In contrast, PrivSTD demonstrates stronger robustness under moderate to relaxed privacy budgets. However, at extremely small privacy budgets (e.g., $\epsilon = 0.5$), PrivSTD performs worse than VDR or AG on some datasets. This behavior reflects a sharp drop in the signal-to-noise ratio in the frequency domain under user-level DP. When each user contributes multiple trajectory points, the noise injected in the frequency coefficients may become comparable to, or even dominate, true low-frequency signals. After inverse DCT, this leads to systematic bias in global density estimates. Since CV-R2R is designed for denoising rather than inversion, it cannot fully correct such front-end bias without access to clean references, resulting in higher error on Cell Count, which is highly sensitive to absolute values. As the privacy budget increases, signal and noise become more separable in the frequency domain, allowing BH-FDR truncation to more reliably preserve dominant structural components. Consequently, enable PrivSTD to quickly surpass baselines and achieve the better performance when ϵ increases.

Results in terms of Hotspot Query. Hotspot Query focuses on preserving local density contrasts and accurately identifying high-density regions, rather than on absolute count precision. Under **user-level DP**, spatial-domain noise injection tends to blur sharp density gradients and weaken hotspot boundaries, and such structural damage is difficult to recover even with larger privacy budgets. As a result, UG, AG, AHP, and MWEM exhibit substantial localization errors. Although VDR alleviates part of the noise through learning-based denoising, its spatial-domain perturbation still weakens hotspot boundaries before the model processes the data, which limits its improvement as the privacy budget increases. By contrast, PrivSTD consistently achieves the lowest localization error under user-level DP. Even at very small privacy budgets, frequency-domain perturbation preserves relative energy distributions to a certain extent, allowing local extrema to remain distinguishable. By suppressing noise-dominated components before reconstruction and leveraging structure-aware recovery, PrivSTD is able to retain sharp density contrasts and accurately recover hotspot regions. This structural advantage enables PrivSTD to maintain stable and superior performance across all privacy budgets.

Results in terms of Forecasting Query. Forecasting Query evaluates the ability to capture smooth temporal evolution and long-term periodic patterns, placing strong emphasis on temporal consistency. Under the **user-level DP** setting, most baseline methods lack explicit temporal modeling and process each time slice independently, causing noise to accumulate over time and weakening forecasting stability. VDR performs particularly poorly on this task: its spatial-domain noise injection disrupts temporal continuity, and

Table 9: Overall Utility Performance of User-level extension (best values are in bold; smaller values are better)

Metric	Method	Tokyo				Stockholm				Austin			
		$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$	$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 3.0$
Cell Count	UG	1.709	1.211	1.110	1.018	0.553	0.527	0.523	0.506	1.078	1.044	0.989	0.963
	AG	0.479	0.474	0.473	0.473	0.423	0.410	0.398	0.396	0.681	0.677	0.659	0.641
	HB-Striped	56.201	28.202	14.203	9.537	13.169	6.691	3.448	2.364	15.772	8.042	4.168	2.878
	AHP	0.587	0.514	0.494	0.485	0.424	0.420	0.422	0.419	0.692	0.698	0.680	0.676
	MWEM	0.495	0.494	0.491	0.488	0.421	0.418	0.414	0.409	0.687	0.679	0.671	0.670
	VDR ²	0.473	0.442	0.440	0.431	0.417	0.408	0.393	0.382	0.679	0.653	0.624	0.620
	PrivBayes	0.575	0.512	0.493	0.483	0.423	0.421	0.420	0.419	0.702	0.698	0.686	0.682
	PrivSTD ¹	0.469	0.433	0.427	0.420	1.134	0.540	0.370	0.328	1.300	0.713	0.572	0.554
Hotspot Query	UG	44.153	41.173	42.340	38.296	32.221	29.285	24.910	23.663	17.736	17.700	17.288	16.926
	AG	6.655	6.552	6.566	6.082	7.761	7.698	7.652	7.606	2.202	2.157	1.995	1.759
	HB-Striped	7.158	7.118	7.111	8.105	8.145	8.137	8.115	7.103	1.457	1.448	1.390	1.376
	AHP	67.341	67.128	66.023	65.088	65.068	47.909	47.613	47.382	35.668	34.804	30.324	26.009
	MWEM	9.256	8.255	7.910	7.326	29.460	28.472	27.733	26.942	15.346	13.742	9.753	6.716
	VDR ²	6.533	6.530	6.342	5.640	7.453	7.415	7.409	7.301	1.565	1.489	1.417	1.327
	PrivBayes	16.247	13.324	11.420	10.736	40.832	29.382	26.472	25.108	6.804	4.924	3.895	3.747
	PrivSTD ¹	6.432	6.356	4.470	4.402	7.261	6.898	5.937	5.559	1.390	1.341	1.311	1.299
Forecasting Query	UG	1.508	1.479	1.440	1.227	1.066	1.043	1.018	0.997	1.403	1.360	1.364	1.348
	AG	1.067	0.975	0.966	0.962	0.986	0.934	0.873	0.869	1.098	0.996	0.950	0.863
	HB-Striped	1.980	1.961	1.924	1.888	1.904	1.820	1.679	1.562	1.831	1.723	1.565	1.451
	AHP	1.274	1.206	1.099	0.767	1.303	1.273	1.216	1.208	1.711	1.655	1.629	1.556
	MWEM	1.207	1.206	1.193	1.191	1.266	1.232	1.229	1.188	1.531	1.514	1.507	0.506
	VDR ²	1.801	1.770	1.747	1.737	1.341	1.250	1.072	0.785	1.232	1.179	0.909	0.762
	PrivBayes	0.905	0.820	0.686	0.765	1.208	1.201	1.200	1.154	1.548	1.532	1.520	1.502
	PrivSTD ¹	0.469	0.433	0.427	0.420	1.360	0.905	0.605	0.510	1.092	0.797	0.672	0.667

subsequent denoising cannot effectively restore long-range structure, resulting in limited improvement even as the privacy budget increases. In addition, our datasets span long time horizons, and the Forecasting Query emphasizes seasonal and long-term patterns rather than short-term trends. This further exposes the limitations of VDR, which was originally designed for short-horizon forecasting. PrivSTD, on the other hand, exhibits strong temporal robustness under user-level DP. Frequency-domain representations naturally capture low-frequency and periodic components, allowing key temporal structures to be partially preserved even under strong noise. As the privacy budget increases, this advantage becomes more pronounced.

9.3 Algorithm Analysis

Algorithm 2 illustrates the overall workflow of PrivSTD. We split the procedure into two parts. The first part applies a frequency noised mechanism, which transforms trajectory density data into the DCT domain, injects Gaussian noise to satisfy (ϵ, δ) -differential privacy, and performs BH-FDR-based truncation to remove noise-dominated high-frequency components, followed by inverse DCT reconstruction to obtain an initial private dataset. The second part applies a denoising model to further reduce DP-induced noise and recover spatiotemporal structure. Since this step operates as post-processing on differentially private outputs, it does not consume additional privacy budget.

Privacy Budget Analysis. The privacy budget of PrivSTD is entirely consumed in Stage 1 by Gaussian noise injection in the frequency domain, with privacy parameters (ϵ, δ) . The other processes are all post-processing operations and therefore incur no

Algorithm 2: PrivSTD

Input: Spatiotemporal density grids X , privacy parameters (ϵ, δ) , FDR level q , denoiser F_θ

Output: Final released data $\hat{X}$

1 Step 1: Frequency Noised Mechanism

- 2 $\Theta_{\text{true}} \leftarrow \text{DCT_TRANSFORM}(X)$;
- 3 $\Theta_{\text{noisy}} \leftarrow \text{GAUSSIAN_NOISE}(\Theta_{\text{true}}, \epsilon, \delta)$;
- 4 $\Theta_{\text{trunc}} \leftarrow \text{BH_FDR_TRUNCATION}(\Theta_{\text{noisy}}, q)$;
- 5 $\tilde{X} \leftarrow \text{INVERSE_DCT}(\Theta_{\text{trunc}})$;

6 Step 2: Denoised (Post-processing)

- 7 $\hat{X} \leftarrow \text{DENOISE}(\tilde{X}; F_\theta)$;
- 8 **return** $\hat{X}$;

additional privacy loss. As a result, PrivSTD satisfies event-level (ϵ, δ) -differential privacy.

Time Complexity Analysis. The computational cost of Stage 1 is dominated by the 2D DCT and IDCT operations over T time slices, with time complexity $O(T \cdot HW \log(HW))$, along with an additional $O(HW \log(HW))$ cost for BH-FDR sorting. The complexity of Stage 2 is mainly determined by the training of the denoising network, where dual sampling introduces only constant-factor overhead for forward and backward propagation, and SPS updates incur negligible additional cost. Overall, the total time complexity is:

$$O(T \cdot HW \log(HW) + \text{TrainCost}(\mathcal{F}_\theta)),$$

where H and W denote the spatial grid dimensions and T denotes the number of temporal slices.

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